

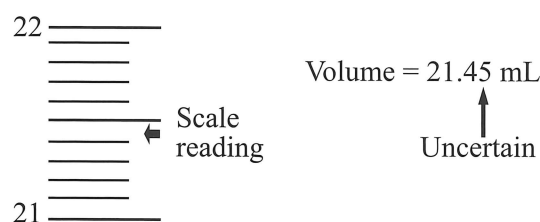
Set 2: Significant figures

Uncertainty and significant figures in measurement

When handling experimental data it is important to consider the appropriate number of significant figures to use and not be tempted to record values with large numbers of significant figures that may be produced by your calculator when processing data. Significant figures in a number are those digits that are known with certainty plus the first digit that is uncertain. All measurements have some level of uncertainty. These arise from the quality of the equipment used, the type of scale and the skill of the experimenter. Some apparatus will have a manufacturer's uncertainty assigned; however, there are cases where you will make a judgement.

Burettes are used in titration experiments to measure the volume of a reactant added to another accurately. A typical 50 mL burette has an analogue scale with a smallest scale division of 0.1 mL. A manufacturer's uncertainty of ± 0.05 mL (representing half the smallest scale division) is often assigned. Depending on your skill level you may judge that you can read the scale on the burette to the nearest 0.01 mL and so reduce the range of uncertainty for each measurement to ± 0.02 mL.

The burette reading in the diagram of 21.45 mL is estimated to the nearest 0.05 mL. The true value lies in a range from 21.40-21.50 mL.



Significant figures in a number are those digits that are known with certainty plus the first digit that is uncertain. In the reading of 21.45 mL the 2, 1 and 4 are known with certainty and the 5 is uncertain. This value has 4 significant figures.

	Number	Number of significant figures
1.	713	3
2.	7.03×10^3	3
3.	11.05	4
4.	0.027	2
5.	9.9643×10^{-7}	5

Whole numbers ending with one or more zeros such as 430 and 500 are ambiguous with respect to the number of significant figures because it is unclear whether the terminating zeros are significant or merely serve to locate the decimal point.

The use of scientific notation avoids ambiguity as shown by the following examples:

1. If 430 has 3 significant figures it is written as 4.30×10^2
2. If 430 has 2 significant figures it is written as 4.3×10^2
3. If 500 has 1 significant figure it is written as 5×10^2

Set 2: Significant figures

Rounding off

Rounding is used to reduce the complexity of a number when it is written with more digits than are wanted or justified. The last digit written should give the best approximation of the number as it was before rounding. If the number before rounding is as close to one number as another, the one ending with an even digit is chosen, zero being regarded as even (example 6).

The following examples illustrate rounding to **three significant figures**.

- | | | | |
|----|---------|-----------|--------|
| 1. | 1.294 | rounds to | 1.29 |
| 2. | 8.12349 | rounds to | 8.12 |
| 3. | 0.01249 | rounds to | 0.0125 |
| 4. | 18.951 | rounds to | 19.0 |
| 5. | 7.1451 | rounds to | 7.15 |
| 6. | 7.145 | rounds to | 7.14 |

Rounding in multi-step calculations

When performing calculations requiring several steps, only round to the appropriate number of significant figures after the final step. This avoids possible errors that can accumulate during a calculation if rounding to the strict number of significant figures is carried out at each step.

Notes

Set 2: Exercises

- How many significant figures are given in each of the following
(a) 454 (c) 100.10 (e) 6000 (g) 14.0
(b) 0.3750 (d) 6.07×10^{-11} (f) 0.0003 (h) 1.103
- Round off the following measurements to three significant figures:
(a) 7.248 (b) 0.017428 (c) 6.275×10^{-3} (d) 11.252×10^7
- Evaluate the following and express the answer to the correct number of significant figures:
(a) $2.65 + 2.5 + 3.61$
(b) $727 + 18.42 - 85.496$
(c) $(2.464 \times 10^{-3}) - (8.7643 \times 10^{-4})$
(d) $8734.2 + (2.81 \times 10^4) - (3.432 \times 10^3)$
- Evaluate the following and quote your answer in exponential notation to the correct number of significant figures.
(a) 7.325×9.27
(b) $(7.81 \times 10^4) \times 0.031 \times 22.4$
(c) $\frac{8.426}{2.98}$
(d) $\frac{3.564 \times 10^6}{7.2538 \times 10^{-2}}$
(e) $\frac{(5.68 \times 10^2) + (2.10 \times 10^{-1})}{2.1 \times 10^{-1}}$
(f) $\frac{(8.23 \times 10^2) - (7.94 \times 10^2)}{8.262 \times 10^{-2}}$
- Calculate the volume of a rectangular block with the dimensions 3.0 cm by 2.0 cm by 1.55 cm.